

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			26	TO BE USED ONLY WHEN INDICATED ON P&ID.
20										810	NIPPLE TO BE CUT FROM SCH 160 PIPE.
20		VA5					***** MODULAR VALVES *****			823	USE OF THIS ITEM IS LIMITED TO MAXIMUM DESIGN TEMPERATURE OF 250°C
30			MODULAR DBB	K01AS02	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
40		VAV	MODULAR DBB	K03AS02	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
50			MODULAR DBB	M01AS02	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		
60		FS1	MODULAR DBB	M03AS02	0.75	2	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
70		VAV	MODULAR DBB	M03ES02	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	1.5	3	FA3	FLAT FACE FLANGES & FULL FACE GASKETS SHALL ONLY BE USED WHEN MATING WITH FLAT FACE EQUIPMENT.
80							***** BALL VALVES *****			FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
80		VB2								FB1	FOR MIST SUPPLY MANIFOLD ONLY
90										FB2	TO BE USED IN FF FLANGE JOINTS IN DIESEL EXHAUST SERVICES
90			BALL	A01AS01	3	4	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO	0	0	FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES
100			BALL	A01AS02	3	4	BALL 150# RF SS A182 F316 316SS TRIM FLOATING METAL RB LO	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
110	U04		BALL	A01AS03	6	6	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT RB GO	0	0	GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117
120	U04		BALL	A01AS04	6	6	BALL 150# RF SS A182 F316 316SS TRIM FLOATING METAL RB GO	0	0	ND9	USE ONLY WITH CONSENT OF MATERIALS ENGINEER.
130			BALL	A01AS07	8	8	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
140			BALL	A01AS08	8	8	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO	0	0	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST
150	FS1		BALL	B01AS01	0.5	3	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO	0	0	UDC	FOR USE WITH PRESSURE CONNECTIONS ONLY
160	FS1		BALL	B01AS02	0.5	3	BALL 150# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO	0	0	UG3	PTFE GASKET ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING
170	U04	VA5	BALL	B01AS03	4	4	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT FB GO	0	0	UG4	BOLTS ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING
180	U04	VA5	BALL	B01AS04	4	4	BALL 150# RF SS A182 F316 316SS TRIM FLOATING METAL FB GO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
190			BALL	B01AS07	6	24	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO	0	0	UJ3	USE 3IN WITH THERMOWELL CONNECTIONS ONLY
200	U04	VA5	BALL	B01AS08	6	24	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO	0	0	UJ4	FOR 2 INCH THERMOWELL CONNECTIONS ONLY
210			BALL	B01JS07	26	42	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A	0	0	UK9	DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.
220			BALL	B01JS08	26	42	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
230	U04	FS1	BALL	B03AS01	0.5	3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO	0	0	VAR	ONLY TO BE USED WHEN SPECIFIED ON THE P&ID.
240	U04	FS1	BALL	B03AS02	0.5	3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
250	FS1		BALL	A03AS01	3	3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
260	FS1		BALL	A03AS02	3	3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL RB LO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
270	FS1		BALL	B03AS07	10	10	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
280							***** BUTTERFLY VALVES *****				
280											
290											
290			BUTTERFLY	T01AS61	10	24	BUTTERFLY 150# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
300			BUTTERFLY	T01JS61	26	42	BUTTERFLY 150# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A	0	0		
310			BUTTERFLY	T03AS61	12	12	BUTTERFLY 300# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
320							***** GATE VALVES *****				
320											
330											
330	U04		GATE	G01AS48	0.5	1.5	GATE 150# RF SS A182 F316 TRIM 16 WEDGE HW	0	0		
340	U04		GATE	G01AS41	2	8	GATE 150# RF SS A351 CF8M TRIM 16 WEDGE HW	0	0		
350	U04	VAR	GATE	G01AS41	10	16	GATE 150# RF SS A351 CF8M TRIM 16 WEDGE HW	0	0		
360	U04	VA5	GATE	G03AS48	0.75	0.75	GATE 300# RF SS A182 F316 TRIM 16 WEDGE HW	0	0		
370	U04	VA5	GATE	G03AS41	2	3	GATE 300# RF SS A351 CF8M TRIM 16 WEDGE HW	0	0		
380							***** GLOBE VALVES *****				
380											
390											
390			GLOBE	L01AS48	0.75	1.5	GLOBE 150# RF SS A182 F316 TRIM 16 OS&Y HW	0	0		
400			GLOBE	L01AS41	2	6	GLOBE 150# RF SS A351 CF8M TRIM 16 OS&Y HW	0	0		
410			GLOBE	L01AS42	8	10	GLOBE 150# RF SS A351 CF8M TRIM 16 OS&Y GO	0	0		
420							***** CHECK VALVES *****				
420											
430											
430		VC1	CHECK	C01AS50	0.75	1.5	CHECK PISTON 150# RF SS A182 F316 TRIM 16	0	0		
440			CHECK	C01AS45	2	6	CHECK DUAL WAFER LUG 150# RF SS A351 CF8M TRIM 16 THROU BOLTED	0	0		
450			CHECK	C01AS52	8	24	CHECK DUAL DBL FLG 150# RF SS A351 CF8M TRIM 16	0	0		
460			CHECK	C01JS52	26	42	CHECK DUAL DBL FLG 150# RF SS A351 CF8M TRIM 16 B16.47 SERIES A	0	0		
470							***** PIPE *****				
470											
480											
480	U04		PIPE	5365140	0.5	1.5	PIPE SCH 80S SMLS SS A312 TP316/316L PE	0	0		
490			PIPE	5365136	1	6	PIPE SCH 40S SMLS SS A312 TP316/316L BE	0	0		
500	U04		PIPE	5365119	8	12	PIPE SCH 30 SMLS SS A312 TP316/316L BE	0	0		
510			PIPE	5365117	14	18	PIPE SCH 10 SMLS SS A312 TP316/316L BE	0	0		
520			PIPE	5365118	20	24	PIPE SCH 20 SMLS SS A312 TP316/316L BE	0	0		
530			PIPE	5365193	26	36	PIPE SCH STD EFW SS A358 316/316L BE	0	0		
540			PIPE	5365193	42	42	PIPE SCH STD EFW SS A358 316/316L BE	0	0		
550	810		PIPE	5365207	0.5	0.5	PIPE SCH 160 SMLS SS A312 TP316/316L PE	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE - ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 535 °C	12/16/2024	BY OZAC	U04	Page 1 of 5	150P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
560			PIPE	5365138	4	24	PIPE SCH 80S SMLS SS A312 TP316/316L-S BE	0	0	
570	U04	UK9	PIPE	5365202	0.5	0.75	PIPE SCH XXS SMLS SS A312 316/316L SS PE	0	0	
580							***** FITTINGS *****			
580										
590										
590			ELBOW	5449033	0.5	1.5	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S	0	0	
600	U04	UK9	ELBOW	5449572	0.5	0.75	ELL 90 DEG LR SCH XXS SS A403 WP316/316L-S	0	0	
610			ELBOW	5449138	1	6	ELL 90 DEG LR SCH 40S SS A403 WP316/316L-S	0	0	
620			ELBOW	5449031	8	12	ELL 90 DEG LR SCH 30 SS A403 WP316/316L-S	0	0	
630			ELBOW	5448890	14	18	ELL 90 DEG LR SCH 10 SS A403 WP316/316L-S	0	0	
640			ELBOW	5449030	20	24	ELL 90 DEG LR SCH 20 SS A403 WP316/316L-S	0	0	
650			ELBOW	5451100	26	42	ELL 90 DEG LR SCH STD 316/316L SS A403 WP316/316L-W	0	0	
660	U04		ELBOW	5449146	0.5	1.5	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S	0	0	
670	U04	UK9	ELBOW	5449143	0.5	0.75	ELL 45 DEG LR XXS SS A403 WP316/316L-S	0	0	
680	U04		ELBOW	5449142	0.5	0.5	ELL BW 45 DEG LR SCH 160 SS A403 WP316/316L-S	0	0	
690			ELBOW	5449144	1	6	ELL 45 SCH 40S SS A403 WP316/316L-S	0	0	
700			ELBOW	5449037	8	12	ELL 45 SCH 30 SS A403 WP316/316L-S	0	0	
710			ELBOW	5448891	14	18	ELL 45 DEG SCH 10 A403 WP316/316L-S	0	0	
720			ELBOW	5449036	20	24	ELL 45 DEG SCH 20 A403 WP316/316L-S	0	0	
730			ELBOW	5451106	26	42	ELL 45 DEG SCH STD 316/316L SS A403 WP316/316L-W	0	0	
740										
740	U04		TEE	5449156	0.5	1.5	TEE SCH 80S SS A403 WP316/316L-S	0	0	
750			TEE	5449147	1	6	TEE SCH 40S SS A403 WP316/316L-S	0	0	
760			TEE	5449043	8	12	TEE SCH 30 SS A403 WP316/316L-S	0	0	
770			TEE	5448892	14	18	TEE SCH 10 SS A403 WP316/316L-S	0	0	
780			TEE	5449039	20	24	TEE SCH 20 SS A403 WP316/316L-S	0	0	
790			TEE	5451109	26	42	TEE SCH STD 316/316L SS A403 WP316/316L-W	0	0	
800	U04	UK9	TEE	5449191	0.5	0.75	TEE SCH XXS SS A403 WP316/316L-S	0	0	
810			TEE RED	5448905	0.5	1	TEE RED SCH 80S SS A403 WP316/316L-S	0.75	1.5	
820			TEE RED	5449250	0.75	1.5	TEE RED SCH 40S/80S SS A403 WP316/316L-S	1	3	
830			TEE RED	5449184	2	4	TEE RED SCH 40S SS A403 WP316/316L-S	3	6	
840			TEE RED	5448889	4	6	TEE RED SCH 30/40S SS A403 WP316/316L-S	8	12	
850			TEE RED	5448880	6	6	TEE RED SCH 10/40S SS A403 WP316/316L-S	14	16	
860			TEE RED	5449053	8	10	TEE RED SCH 30 SS A403 WP316/316L-S	10	12	
870			TEE RED	5448896	8	12	TEE RED SCH 10/30 SS A403 WP316/316L-S	14	18	
880			TEE RED	5449167	8	12	TEE RED SCH 20/30 SS A403 WP316/316L-S	20	24	
890			TEE RED	5448898	14	16	TEE RED SCH 10 SS A403 316/316L-S	16	18	
900			TEE RED	5449168	14	18	TEE RED SCH 20/10 SS A403 WP316/316L-S	20	24	
910			TEE RED	5449041	20	20	TEE RED SCH 20 SS A403 WP316/316L-S	24	24	
920			TEE RED	5449169	24	24	TEE RED SCH STD/20 SS A403 WP316/316L-W	26	26	
930			TEE RED	5449042	26	36	TEE RED SCH STD SS A403 WP316/316L-W	28	42	
940										
940	U04		CAP	5449049	0.5	1.5	CAP SCH 80S SS A403 WP316/316L-S	0	0	
950			CAP	5449148	2	6	CAP SCH 40S SS A403 WP316/316L-S	0	0	
960			CAP	5449044	8	12	CAP SCH 30 SS A403 WP316/316L-S	0	0	
970			CAP	5448893	14	18	CAP SCH 10 SS ASTM A403 WP316/316L-S	0	0	
980			CAP	5449040	20	24	CAP SCH 20 SS A403 WP316/316L-S	0	0	
990			CAP	5449256	26	42	CAP SCH STD SS A403 WP316/316L-S	0	0	
1000										
1000	U04		RED CONC	5449158	0.5	1	REDUCER CONC SCH 80S SS A403 WP316/316L-S	0.75	1.5	
1010			RED CONC	5549110	0.75	1.5	REDUCER CONC SCH 40S/80S SS A403 WP316/316L-S	2	4	
1020			RED CONC	5449149	0.5	5	REDUCER CONC SCH 40S SS A403 WP316/316L-S	0.75	6	
1030			RED CONC	5448899	4	6	REDUCER CONC SCH 30/40S SS A403 WP316/316L-S	8	12	
1040			RED CONC	5448881	6	6	REDUCER CONC SCH 10/40S SS A403 WP316/316L-S	14	14	
1050			RED CONC	5448900	8	10	REDUCER CONC SCH 30 SS A403 WP316/316L-S	10	12	
1060			RED CONC	5448901	8	12	REDUCER CONC SCH 10/30 SS A403 WP316/316L-S	14	18	
1070			RED CONC	5449173	12	12	REDUCER CONC SCH 20/30 SS A403 WP316/316L-S	20	20	
1080			RED CONC	5448894	14	16	REDUCER CONC SCH 10 SS A403 WP316/316L-S	16	18	
1090			RED CONC	5450720	18	18	REDUCER CONC SCH STD/10 SS A403 WP316/316L-W	26	28	
1100			RED CONC	5449174	14	18	REDUCER CONC SCH 20/10 SS A403 WP316/316L-S	20	24	
1110			RED CONC	5449175	20	20	REDUCER CONC SCH 20 SS A403 WP316/316L-S	24	24	
1120			RED CONC	5450721	20	24	REDUCER CONC SCH STD/20 SS A403 WP316/316L-W	26	36	
1130			RED CONC	5450722	26	36	REDUCER CONC SCH STD SS A403 WP316/316L-W	28	42	
1140	U04		RED ECC	5449159	0.5	1	REDUCER ECC SCH 80S SS A403 WP316/316L-S	0.75	1.5	
1150			RED ECC	5549310	0.75	1.5	REDUCER ECC SCH 40S/80S SS A403 WP316/316L-S	2	4	
1160			RED ECC	5449150	0.75	5	REDUCER ECC SCH 40S SS A403 WP316/316L-S	1	6	
1170			RED ECC	5449151	4	6	REDUCER ECC SCH 30/40S SS A403 WP316/316L-S	8	12	
1180			RED ECC	5448882	6	6	REDUCER ECC SCH 10/40S SS A403 WP316/316L-S	14	14	
1190			RED ECC	5449152	8	10	REDUCER ECC SCH 30 SS A403 WP316/316L-S	10	12	
1200			RED ECC	5449153	8	12	REDUCER ECC SCH 10/30 SS A403 WP316/316L-S	14	18	
1210			RED ECC	5449170	12	12	REDUCER ECC SCH 20/30 SS A403 WP316/316L-S	20	20	
1220			RED ECC	5448895	14	16	REDUCER ECC SCH 10 SS A403 WP316/316L-S	16	18	
1230			RED ECC	5449171	14	18	REDUCER ECC SCH 20/10 SS A403 WP316/316L-S	20	24	

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		ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE - ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 535 °C	12/16/2024	BY OZAC	U04	Page 2 of 5	150P22

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1240			RED ECC	5450723	18	18	REDUCER ECC SCH STD/10 SS A403 WP316/316L-W	26	28					
1250			RED ECC	5449172	20	20	REDUCER ECC SCH 20 SS A403 WP316/316L-S	24	24					
1260			RED ECC	5450724	20	24	REDUCER ECC SCH STD/20 SS A403 WP316/316L-W	26	36					
1270			RED ECC	5450725	26	36	REDUCER ECC SCH STD SS A403 WP316/316L-W	28	42					
1280														
1280			OLETS	5504311	0.75	1.5	NIPOFLANGE 150# RF SCH 40S/80S SS A182 F316/316L	2	6					
1290			OLETS	5504331	0.75	1.5	NIPOFLANGE 150# RF SCH 30/80S SS A182 F316/316L	8	12					
1300			OLETS	5504312	0.75	1.5	NIPOFLANGE 150# RF SCH 10/80S SS A182 F316/316L	14	18					
1310			OLETS	5504334	0.75	1.5	NIPOFLANGE 150# RF SCH 20/80S SS A182 F316/316L	20	24					
1320			OLETS	5504343	0.75	1.5	NIPOFLANGE 150# RF SCH STD/80S SS A182 F316/316L	26	42					
1330			OLETS	5504011	2	2	NIPOFLANGE 150# RF SCH 40S/40S SS A182 F316/316L	3	6					
1340			OLETS	5504031	2	2	NIPOFLANGE 150# RF SCH 30/40S SS A182 F316/316L	8	12					
1350			OLETS	5504012	2	2	NIPOFLANGE 150# RF SCH 10/40S SS A182 F316/316L	14	18					
1360			OLETS	5504034	2	2	NIPOFLANGE 150# RF SCH 20/40S SS A182 F316/316L	20	24					
1370			OLETS	5504043	2	2	NIPOFLANGE 150# RF SCH STD/40S SS A182 F316/316L	26	42					
1380			OLETS	5504412	1	1	NIPOFLANGE 300# RF SCH 80S SS A182 F316/316L	1.5	1.5					
1390			OLETS	5504411	0.75	1.5	NIPOFLANGE 300# RF SCH 40S/80S SS A182 F316/316L	2	6					
1400		UJ2	OLETS	5504132	0.75	1.5	NIPOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4	6					
1410		UJ2	OLETS	5504145	0.75	1.5	NIPOFLANGE 300# RF SCH 30/160 SS A182 F316/316L	8	12					
1420		UJ2	OLETS	5504146	0.75	1.5	NIPOFLANGE 300# RF SCH 10/160 SS A182 F316/316L	14	18					
1430		UJ2	OLETS	5504147	0.75	1.5	NIPOFLANGE 300# RF SCH 20/160 SS A182 F316/316L	20	24					
1440		UJ2	OLETS	5504144	0.75	1.5	NIPOFLANGE 300# RF SCH STD/160 SS A182 F316/316L	26	42					
1450			OLETS	5504528	3	4	NIPOFLANGE 300# RF SCH 20/160 SS A182 F316/316L	20	20					
1460	U04	UJ4	OLETS	5504143	2	2	NIPOFLANGE 300# RF SCH 20/160 SS A182 F316/316L	20	20					
1470	U04			5504422	1.5	1.5	NIPOFLANGE 300# RF SCH 20/80S SS A182 F316/316L	20	20					
1480	U04	UJ4	OLETS	5504148	2	2	NIPOFLANGE 300# RF SCH 30/160 SS A182 F316/316L	10	10					
1490			OLETS	5504414	0.75	1.5	NIPOFLANGE 300# RF SCH 30/80S SS A182 F316/316L	8	8					
1500	U04		OLETS	5504032	2	2	NIPOFLANGE 300# RF SCH 40S/40S SS A182 F316/316L	3	3					
1510	U04	UJ4	OLETS	5504151	2	2	NIPOFLANGE 300# RF SCH STD/160 SS A182 F316/316L	36	36					
1520	U04	UJ3	OLETS	5504529	3	3	NIPOFLANGE 300# RF SCH STD/160 SS A182 F316/316L	36	36					
1530	U04	FF5	OLETS	5504523	0.75	1	NIPOFLANGE FLAT BOTTOM 150# RF SCH 80S SS A182 F316/316L	0	0					
1540														
1540			OLETS	5510354	0.5	1.5	WELDOLET SCH 40S/80S SS A182 F316/316L	2	6					
1550			OLETS	5510355	0.75	1.5	WELDOLET SCH 30/80S SS A182 F316/316L	8	12					
1560			OLETS	5510356	0.75	1.5	WELDOLET SCH 10/80S SS A182 F316/316L	14	18					
1570			OLETS	5510357	0.75	1.5	WELDOLET SCH 20/80S SS A182 F316/316L	20	24					
1580			OLETS	5510358	0.75	1.5	WELDOLET STD/80S SS A182 F316/316L	26	42					
1590			OLETS	5509631	2	3	WELDOLET SCH 40S/40S SS A182 F316/316L	4	6					
1600			OLETS	5509634	2	4	WELDOLET SCH 30/40S SS A182 F316/316L	8	12					
1610			OLETS	5509630	2	6	WELDOLET SCH 10/40S SS A182 F316/316L	14	18					
1620			OLETS	5509633	2	6	WELDOLET SCH 20/40S SS A182 F316/316L	20	24					
1630			OLETS	5509650	2	6	WELDOLET STD/40S SS A182 F316/316L	26	42					
1640			OLETS	5509641	8	8	WELDOLET SCH 20/30 SS A182 F316/316L	24	24					
1650			OLETS	5509642	8	12	WELDOLET STD/30 SS A182 F316/316L	26	42					
1660	U04		OLETS	5504375	0.75	1	WELDOLET FLAT BOTTOM SCH 80S SS A182 F316/F316L	0	0					
1670														
1670	U04	UD7	OLETS	5515771	0.75	0.75	THREDOLET 3000# SS A182 F316/316L	3	4					
1680														
1680		UJ2	ELBOFLG	5516152	1.5	1.5	ELBOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4	6					
1690		UJ2	ELBOFLG	5516153	1.5	1.5	ELBOFLANGE 300# RF SCH 30/160 SS A182 316/316L	8	12					
1700		UJ2	ELBOFLG	5516154	1.5	1.5	ELBOFLANGE 300# RF SCH 10/160 SS A182 316/316L	14	18					
1710		UJ2	ELBOFLG	5516155	1.5	1.5	ELBOFLANGE 300# RF SCH 20/160 SS A182 316/316L	20	24					
1720		UJ2	ELBOFLG	5516157	1.5	1.5	ELBOFLANGE 300# RF STD/160 SS A182 316/316L	26	42					
1730														
1730	U04	ND9	Coupling	5480142	1.5	1.5	COUPLING 3000# SW SS A182 F316/316L	0	0					
1740														
1740	U04	UDC	FITTING SCRDR	5498383	0.5	0.5	PLUG ROUND HEAD SCRDR SS A182 F316/316L	0	0					
1750														
1750		FS1					***** FLANGES *****							
1760														
1760	U04	FA8	FLANGE	5600655	0.5	0.75	FLG SCRDR (FNPT) REDUCING 150# RF SS A182 F316/316L	0.75	2					
1770	U04	FA8	FLANGE	5600656	0.5	0.5	FLG SCRDR (FNPT) REDUCING 300# RF SS A182 F316/316L	0.75	2					
1780	U04	FA8	FLANGE	5600657	1.5	1.5	FLG SCRDR (FNPT) 150# RF SS A182 F316/316L	0	0					
1790	U04		FLANGE	5556281	2	2	FLG WN REDUCING 150# RF A182 F316/316L BORED TO SUIT SCH 40S	3	3					
1800														
1800		SB6	FLANGE	5851200	0.5	2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE	0	0					
1810	U04		FLG ADAPTER	5851209	0.75	1	FLANGE ADAPTER 150# RF 316 SS X 25 MM TUBE	0	0					
1820	U04		FLG ADAPTER	5851203	0.5	2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0					
1830	U04		FLG ADAPTER	5851217	1.5	1.5	FLANGE ADAPTER 150# RF 316 SS X 38 MM TUBE	0	0					
1840														
1840			FLANGE	5582724	0.5	1.5	FLG WN 150# RF SS A182 F316/316L SCH 80S BORE	0	0					
1850			FLANGE	5582725	0.75	6	FLG WN 150# RF SS A182 F316/316L SCH 40S BORE	0	0					
1860	U04		FLANGE	5582726	8	12	FLG WN 150# RF SS A182 F316/316L SCH 30 BORE	0	0					
	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
			ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 1 MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE - ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 535 °C		12/16/2024	BY OZAC	U04	Page 3 of 5	150P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1870			FLANGE	5582722	14	18	FLG WN 150# RF SS A182 F316/316L SCH 10 BORE	0	0	
1880			FLANGE	5582723	20	24	FLG WN 150# RF SS A182 F316/316L SCH 20 BORE	0	0	
1890			FLANGE	5583005	26	42	FLG WN 150# RF SS A182 F316/316L STD B16.47 SERIES A	0	0	
1900		FA3	FLANGE	5582729	2	6	FLG WN 150# FF SS A182 F316/316L SCH 40S BORE	0	0	
1910	U04	FA3	FLANGE	5582735	18	18	FLG WN 150# FF SS A182 F316/316L SCH 10 BORE	0	0	
1920	U04	FA3	FLANGE	5582734	24	24	FLG WN 150# FF SS A182 F316/316L SCH 20 BORE	0	0	
1930	U04	FA3	FLANGE	5582732	8	10	FLG WN 150# FF SS A182 F316/316L SCH 30 BORE	0	0	
1940	U04			5884164	0.5	0.5	FLG WN 150# RF SS A182 F316/316L SCH SCH 160	0	0	
1950			FLANGE	5582991	0.5	1.5	FLG WN 300# RF SS A182 F316/316L SCH 80S BORE	0	0	
1960			FLANGE	5582995	2	6	FLG WN 300# RF SS A182 F316/316L SCH 40S BORE	0	0	
1970			FLANGE	5582996	8	12	FLG WN 300# RF SS A182 F316/316L SCH 30 BORE	0	0	
1980			FLANGE	5582993	14	18	FLG WN 300# RF SS A182 F316 SCH 10S BORE	0	0	
1990			FLANGE	5582992	20	24	FLG WN 300# RF SS A182 F316/316L SCH 20 BORE	0	0	
2000			FLANGE	5583010	26	42	FLG WN 300# RF SS A182 F316/316L STD B16.47 SERIES A	0	0	
2010			FLANGE	5583535	2	6	FLG WN 600# RF SS A182 F316/316L SCH 40S BORE	0	0	
2020	U04		FLANGE	5582998	0.5	1.5	FLG WN 600# RF SS A182 F316/316L SCH 80S BORE	0	0	
2030	U04		FLANGE	5583021	3	6	FLG WN 900# RF SS A182 F316/316L SCH 40S BORE	0	0	
2040			FLANGE	5583979	1.5	1.5	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE	0	0	
2050			FLANGE	5583970	2	2	FLG WN 1500# RF SS A182 F316/316L SCH 40S BORE	0	0	
2060			BLIND	5616520	0.5	24	FLG BLIND 150# RF SS A182 F316/316L	0	0	
2070		FA8	BLIND	5616521	26	42	FLG BLIND 150# RF SS A182 F316/316L B16.47 SERIES A	0	0	
2080			BLIND	5616529	0.5	24	FLG BLIND 300# RF SS A182 F316/316L	0	0	
2090		FS1	BLIND	5616530	26	42	FLG BLIND 300# RF SS A182 F316/316L B16.47 SERIES A	0	0	
2100										
2100							***** ORIFICE FLANGES *****			
2110										
2110			ORIFICE FLG	5640598	2	6	WN ORIF SET 300# RF SS A182 F316/316L SCH 40S BORE	0	0	
2120			ORIFICE FLG	5640597	8	12	WN ORIF SET 300# RF SS A182 F316/316L SCH 30 BORE WN	0	0	
2130			ORIFICE FLG	5640596	14	18	WN ORIF SET 300# RF SS A182 F316/316L SCH 10 BORE WN	0	0	
2140			ORIFICE FLG	5640595	20	24	WN ORIF SET 300# RF SS A182 F316/316L SCH 20 BORE WN	0	0	
2150			ORIFICE FLG	5640610	26	42	WN ORIF SET 300# RF SS A182 F316/316L STD WT B16.47 SERIES A	0	0	
2160										
2160							***** PIPE NIPPLES *****			
2170										
2170		810		0			NIPPLE PBE CUT FROM PIPE			
2180	U04	FB1	NIPPLE	5542164	0.75	0.75	NIPPLE SCH 160/80S A312 TP316/316L TOE/POE 3 IN LONG	0	0	
2190										
2190		FS1					***** GASKETS *****			
2200										
2200			GASKET	5673043	0.5	24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2210			GASKET	5674011	26	42	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
2220	U04	FA3	GASKET	5664562	0.75	24	GASKET 150# FF RING EPDM 3MM THK	0	0	
2230	U04	FB2		5664568	2	24	GASKET 150# FF RING MICA WITH TANGED INSERT 3MM THK	0	0	
2240	U04	UG3	GASKET	5664201	2	24	GASKET 150# RF FLAT RING PTFE	0	0	
2250	U04	FB2		5664569	2	24	GASKET 150# RF RING MICA WITH TANGED INSERT 3MM THK	0	0	
2260			GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2270			GASKET	5674032	26	42	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
2280	U04	GA6	GASKET	5673017	2	3	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
2290			GASKET	5673103	0.5	6	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2300	U04		GASKET	5673117	3	6	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2310	U04		GASKET	5672251	1.5	1.5	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2320										
2320		B1					***** BOLTING *****			
2330										
2330	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
2340	U04	UG4	BOLTING	5675768	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS	0	0	
2350	U04	26	BOLTING	SP-2745	6	6	STUD-BOLT WITH 2 NUTS API -6A718 M20 X 115 MM	0	0	
2360	U04	26	BOLTING	SP-2746	10	10	STUD-BOLT WITH 2 NUTS API-6A718 M24 X 135 MM	0	0	
2370										
2370							***** BLINDS & SPACERS *****			
2380										
2380			SPEC BLIND	5750421	0.75	14	SPECTACLE BLIND 150# RF SS A240 GR 316	0	0	
2390			SPEC BLIND	5750422	0.75	10	SPECTACLE BLIND 300# RF SS A240 GR 316	0	0	
2400										
2400			PADDLE	5750521	16	24	PADDLE BLANK 150# RF SS A240 GR 316	0	0	
2410		SB6	PADDLE	5750526	26	42	PADDLE BLANK 150# RF SS A240 GR 316 B16.47 SERIES A	0	0	
2420			PADDLE	5750522	12	24	PADDLE BLANK 300# RF SS A240 GR 316	0	0	
2430			PADDLE	5750527	26	36	PADDLE BLANK 300# RF SS A240 GR 316 B16.47 SERIES A	0	0	
2440			PADDLE	5750621	16	24	PADDLE SPACER 150# RF SS A240 GR 316	0	0	
2450		SB6	PADDLE	5750626	26	42	PADDLE SPACER 150# RF SS A240 GR 316 B16.47 SERIES A	0	0	
2460			PADDLE	5750622	12	24	PADDLE SPACER 300# RF SS A240 GR 316	0	0	
2470			PADDLE	5750627	26	36	PADDLE SPACER 300# RF SS A240 GR 316 B16.47 SERIES A	0	0	
2480										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE - ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 535 °C	12/16/2024	BY OZAC	U04	Page 4 of 5	150P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
2490							*****MISCELLANEOUS*****			
2490										
2500										
2500	U04	823	FLG GUARD	5558881	0.5	10	SAFETY SHIELD PTFE SUITABLE FOR 150# FLANGE	0	0	
2510	U04		FLG GUARD	5558882	0.5	6	SAFETY SHIELD PTFE SUITABLE FOR 300# FLANGE	0	0	

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